



quiz.vu.edu.pk/QuizQuestion

5



MC170200830: Kainat Ashfaq

Time Left 83
sec(s)

MTH634:Quiz No. 2

Quiz Start Time: 02:39 PM, 14 August 2018

Question # 2 of 4 (Start time: 02:41:37 PM, 14 August 2018)

Total Marks: 1

Every discrete space is not a regular space.

Select the correct option



True



False

Click to Save Answer & Move to Next Question



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5



MC170200830: Kainat Ashfaq

Time Left 58 sec(s)

MTH634: Quiz No. 2

Quiz Start Time: 02:39 PM, 14 August 2018

Question # 4 of 4 (Start time: 02:43:18 PM, 14 August 2018)

Total Marks: 1

Which of the following statement is false?

Select the correct option



A set X with indiscrete topology is compact.



A set X with any topology containing finite number subsets of X is compact.



A finite set X with any topology is compact.



An infinite set X with discrete topology is compact.

[Click to Save Answer & Move to Next Question](#)



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5



MC170200830: Kainat Ashfaq

Time Left 17 sec(s)

MTH634:Quiz No. 2

Quiz Start Time: 02:39 PM, 14 August 2018

Question # 1 of 4 (Start time: 02:39:20 PM, 14 August 2018)

Total Marks: 1

If (X, τ) be a compact Hausdorff space then (X, τ) is not normal.

Select the correct option

Reload Math Equations

True



False



Click to Save Answer & Move to Next Question



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2



MC170201745: Shakeel Ahmad

Time Left 18 sec(s)

MTH634:Quiz No. 2

Quiz Start Time: 09:29 AM, 14 August 2018

Question # 3 of 4 (Start time: 09:31:13 AM, 14 August 2018)

Total Marks: 1

A topological space X is called a connected space iff there exists a pair of subsets of X both nonempty and both open and closed.

Select the correct option

Reload Math Equations

True



False



Click to Save Answer & Move to Next Question



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2



MC170201745: Shakeel Ahmad

Time Left 42 sec(s)

MTH634:Quiz No. 2

Quiz Start Time: 09:29 AM, 14 August 2018

Question # 2 of 4 (Start time: 09:30:19 AM, 14 August 2018)

Total Marks: 1

Which of the following statement is false?

Select the correct option



Every metric space is second countable.



Every metric space is Hausdorff space.



Every metric space is normal space.



none



Click to Save Answer & Move to Next Question



quiz.vu.edu.pk/QuizQuestion.

2



MC170201745: Shakeel Ahmad

Time Left 72 sec(s)

MTH634:Quiz No. 2

Quiz Start Time: 09:29 AM, 14 August 2018

Question # 1 of 4 (Start time: 09:29:56 AM, 14 August 2018)

Total Marks: 1

Consider $\mathbb{R}$ with usual topology. Then $A = (0, 1) \cup (3, 5]$ is disconnected subset of $\mathbb{R}$.

Select the correct option

Reload Math Equations

True



False



Click to Save Answer & Move to Next Question



quiz.vu.edu.pk/QuizQuestion.

2



MC170201745: Shakeel Ahmad

Time Left

78

sec(s)

MTH634:Quiz No. 2

Quiz Start Time: 09:29 AM, 14 August 2018

Question # 4 of 4 (Start time: 09:32:30 AM, 14 August 2018)

Total Marks: 1

An open interval in

 $\mathbb{R}$

with usual topology is not compact.

Select the correct option

Reload Math Equations

True



False



Click to Save Answer & Move to Next Question

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2



MC170202458: Aneeta Shaheen

Time Left 81 sec(s) 

MTH634:Quiz No. 2

Quiz Start Time: 10:54 AM, 14 August 2018

Question # 2 of 4 (Start time: 10:54:27 AM, 14 August 2018)

Total Marks: 1

Consider $\mathbb{R}$ with usual topology. There exists no homomorphism between an open interval of $\mathbb{R}$ and a closed interval of $\mathbb{R}$.

Select the correct option

 Reload Math Equations

True



False

[Click to Save Answer & Move to Next Question](#)

Which of the following statement is false?

Select the correct option



Every metric space is second countable.



Every metric space is Hausdorff space.



Every metric space is normal space.



none

Click to Save Answer & Move to Next Question

 quiz.vu.edu.pk/QuizQuestion.

2



MC170202458: Aneeta Shaheen

Time Left 76 sec(s) 

MTH634:Quiz No. 2

Quiz Start Time: 10:54 AM, 14 August 2018

Question # 3 of 4 (Start time: 10:55:29 AM, 14 August 2018)

Total Marks: 1

Every compact subspace of a Hausdorff space is not closed.

Select the correct option



True



False

[Click to Save Answer & Move to Next Question](#)



quiz.vu.edu.pk/QuizQuestion.

2



MC170201745: Shakeel Ahmad

Time Left

78

sec(s)

MTH634:Quiz No. 2

Quiz Start Time: 09:29 AM, 14 August 2018

Question # 4 of 4 (Start time: 09:32:30 AM, 14 August 2018)

Total Marks: 1

An open interval in

 $\mathbb{R}$

with usual topology is not compact.

Select the correct option

Reload Math Equations

True



False



Click to Save Answer & Move to Next Question

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2



MC170201907: Nadia Sadaf

MTH634:Quiz No. 2

Quiz Start Time: 12

Question # 2 of 4 (Start time: 12:44:39 PM, 14 August 2018)

All the metric spaces are not normal spaces.

Select the correct option



True



False

Click to Save Answer & Move

quiz.vu.edu.pk/QuizQuestion.

2



MC170201907: Nadia Sadaf

Time Left 85 sec(s)

MTH634:Quiz No. 2

Quiz Start Time: 12:44 PM, 14 August 2018

Question # 3 of 4 (Start time: 12:46:43 PM, 14 August 2018)

Total Marks: 1

$\mathbb{R}$ with usual topology is a T_4 - space.

Select the correct option

Reload Math Equations

☒	True
☐	False

Click to Save Answer & Move to Next Question

quiz.vu.edu.pk/QuizQuestion.

2



MC170201907: Nadia Sadaf

Time Left 60 sec(s)

MTH634:Quiz No. 2

Quiz Start Time: 12:44 PM, 14 August 2018

Question # 4 of 4 (Start time: 12:47:06 PM, 14 August 2018)

Total Marks: 1

Which of the following statement is true?

Select the correct option

Reload Math Equations

☐	$\mathbb{R}$ with usual topology is compact.
☒	$\mathbb{R}$ with indiscrete topology is compact.
☐	$\mathbb{R}$ with discrete topology is compact.
☐	$\mathbb{R}$ with cofinite topology is not compact.

Click to Save Answer & Move to Next Question

quiz.vu.edu.pk/QuizQuestion.

2



MC170201907: Nadia Sadaf

Time Left 60 sec(s)

MTH634:Quiz No. 2

Quiz Start Time: 12:44 PM, 14 August 2018

Question # 1 of 4 (Start time: 12:44:03 PM, 14 August 2018)

Total Marks: 1

A topological space X is called a connected space iff there exists a pair of subsets of X both nonempty and both open and closed.

Select the correct option

Reload Math Equations

☐	True
☒	False

Click to Save Answer & Move to Next Question

The group G is a copy of the group H , if the function

$$f : G \rightarrow H$$

is -----

Select the correct option

☐ homomorphic

☐ isomorphic

☐ onto

☐ bijective

AHMAD MSC(MATH)
Areeb2113@gmail.com

Click to Save

AHMAD MSC(MATH)
Areeb2113@gmail.com

Time Left 58 sec(s)

Quiz Start Time: 06:58 PM, 18 August 2018

Question # 2 of 10 (Start time: 06:58:47 PM, 18 August 2018) Total Marks: 1

If N is a normal subgroup of $(G, \cdot)$, the set of cosets $G/N = \{Ng | g \in G\}$ forms a group $(G/N, \cdot)$, where the operation is defined by $(Ng1) \cdot (Ng2) = N(g1 \cdot g2)$. This group is called the quotient group or factor group of G by N .

Select the correct option

- | | |
|----------------------------------|-------|
| ☒ | True |
| ☐ | False |

Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
Areeb2113@gmail.com

Time Left 28 sec(s)

Quiz Start Time: 06:58 PM, 18 August 2018

Question # 3 of 10 (Start time: 06:59:36 PM, 18 August 2018)

Total Marks: 1

The quotient group R/Z is isomorphic to the circle group

$$W = \{ e^{i\theta} \in \mathbb{C} \mid \theta \in \mathbb{R} \}.$$

Select the correct option

Reload Math Equations

☒	True
☐	False

Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
Areeb2113@gmail.com

Time Left 74 sec(s)

Quiz Start Time: 06:58 PM, 18 August 2018

Question # 4 of 10 (Start time: 07:00:44 PM, 18 August 2018)

Total Marks: 1

The converse of Lagrange's theorem holds for abelian groups.

Select the correct option

- ☒ True
- ☐ False

Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
Areeb2113@gmail.com

Time Left 26 sec(s)

Quiz Start Time: 06:58 PM, 18 August 2018

Question # 5 of 10 (Start time: 07:01:04 PM, 18 August 2018)

Total Marks: 1

Let ϕ be a homomorphism of a group G into a group G' .
If H is a subgroup of G , then $\phi[H]$ is a subgroup of G' .

Select the correct option

Reload Math Equations

☒	True
☐	False

Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
Areeb21113@gmail.com

Time Left 75 sec(s)

Quiz Start Time: 06:58 PM, 18 August 2018

Question # 6 of 10 (Start time: 07:02:16 PM, 18 August 2018)

Total Marks: 1

Let $(G, \cdot)$ be a group with subgroup H . For $a, b \in G$,
we say $a \equiv b \pmod H$ if and only if $ab^{-1} \in H$.

Select the correct option

Reload Math Equations

☒	True
☐	False

Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
Areeb2113@gmail.com

Time Left 81 sec(s)

Quiz Start Time: 06:58 PM, 18 August 2018

Question # 7 of 10 (Start time: 07:02:35 PM, 18 August 2018) Total Marks: 1

The relation

$$a \equiv b$$

mod H is an equivalence relation on G.

Select the correct option

Reload Math Equations

☒	True
☐	False

Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
Areeb2113@gmail.com

Time Left 73 sec(s)

Quiz Start Time: 06:58 PM, 18 August 2018

Question # 8 of 10 (Start time: 07:02:47 PM, 18 August 2018) Total Marks: 1

If N is a normal subgroup of a group G , the left cosets of N in G are the same as the right cosets of N in G .

Select the correct option

- ☒ True
- ☐ False

Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
Areeb2113@gmail.com

Time Left 71 sec(s)

Quiz Start Time: 06:58 PM, 18 August 2018

Question # 9 of 10 (Start time: 07:03:07 PM, 18 August 2018) Total Marks: 1

If N is a normal subgroup of a group G , the left cosets of N in G are not the same as the right cosets of N in G

Select the correct option

- ☒ False
- ☐ True

Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
Areeb2113@gmail.com

Time Left 82 sec(s)

Quiz Start Time: 06:58 PM, 18 August 2018

Question # 10 of 10 (Start time: 07:03:31 PM, 18 August 2018)

Total Marks: 1

Let

$$\phi : G \rightarrow H$$

be a homomorphism, then $\phi(a^{-1}) = \phi(a)^{-1}$ for $a \in G$.

Select the correct option

Reload Math Equations

True



False



Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
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Time Left 65 sec(s)

Quiz Start Time: 07:19 PM, 18 August 2018

Question # 3 of 10 (Start time: 07:19:31 PM, 18 August 2018) Total Marks: 1

$(\mathbb{Z}_n, +)$ is a cyclic group with 1 as a generator.

Select the correct option

Reload Math Equations

☒	True
☐	False

Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
Areeb2113@gmail.com

Time Left 83 sec(s)

Quiz Start Time: 07:19 PM, 18 August 2018

Question # 4 of 10 (Start time: 07:19:59 PM, 18 August 2018) Total Marks: 1

The group G is a copy of the group H, if the function

$$f : G \rightarrow H$$

is -----

Select the correct option

Reload Math Equations

- ☐ homomorphic
- ☒ isomorphic
- ☐ onto
- ☐ bijective

Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
Areeb2113@gmail.com

Time Left 77 sec(s)

Quiz Start Time: 07:19 PM, 18 August 2018

Question # 5 of 10 (Start time: 07:20:10 PM, 18 August 2018) Total Marks: 1

Let H be a subgroup of a group G . Then $(aH)(bH) = a(bH)$ is not always true when G is non-abelian.

Select the correct option

- | | |
|----------------------------------|-------|
| ☐ | False |
| ☒ | True |

Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
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Time Left 79 sec(s)

Quiz Start Time: 07:19 PM, 18 August 2018

Question # 6 of 10 (Start time: 07:20:27 PM, 18 August 2018) Total Marks: 1

Any subgroups of Abelian group G are not normal subgroups of G .

Select the correct option

- ☒ False
- ☐ True

Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
Areeb2113@gmail.com

Time Left 81 sec(s)

Quiz Start Time: 07:19 PM, 18 August 2018

Question # 7 of 10 (Start time: 07:20:46 PM, 18 August 2018)

Total Marks: 1

$Hg = gH$, for all $g \in G$, if and only if H is a normal subgroup of G .

Select the correct option

Reload Math Equations

True



False



Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
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Time Left 79 sec(s)

Quiz Start Time: 07:19 PM, 18 August 2018

Question # 9 of 10 (Start time: 07:21:08 PM, 18 August 2018) Total Marks: 1

Let

$$\phi : G \rightarrow H$$

be an isomorphism, then if G is abelian, then H is not abelian.

Select the correct option

Reload Math Equations

☐	True
☒	False

Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
Areeb2113@gmail.com

Time Left 79 sec(s)

Quiz Start Time: 07:19 PM, 18 August 2018

Question # 10 of 10 (Start time: 07:21:23 PM, 18 August 2018) Total Marks: 1

If N is a normal subgroup of a group G , the left cosets of N in G are the same as the right cosets of N in G .

Select the correct option

- ☒ True
- ☐ False

Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
Areeb2113@gmail.com

Time Left 80 sec(s)

Quiz Start Time: 07:31 PM, 18 August 2018

Question # 3 of 10 (Start time: 07:32:09 PM, 18 August 2018)

Total Marks: 1

Any subgroups of Abelian group G are not normal subgroups of G .

Select the correct option

- ☒ False
- ☐ True

Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
Areeb2113@gmail.com

Time Left 77 sec(s)

Quiz Start Time: 07:31 PM, 18 August 2018

Question # 4 of 10 (Start time: 07:32:26 PM, 18 August 2018) Total Marks: 1

Let ϕ be a homomorphism of a group G into a group G' .
If K' is a subgroup of G' , then $\phi^{-1}[K']$ is a subgroup of G .

Select the correct option

Reload Math Equations

☒	True
☐	False

Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
Areeb2113@gmail.com

Time Left 77 sec(s)

Quiz Start Time: 07:31 PM, 18 August 2018

Question # 6 of 10 (Start time: 07:32:51 PM, 18 August 2018) Total Marks: 1

Let

$$\phi : G \rightarrow H$$

be an isomorphism, then if G is abelian, then H is not abelian.

Select the correct option

Reload Math Equations

☐	True
☒	False

Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
Areeb2113@gmail.com

Time Left 55 sec(s)

Quiz Start Time: 07:48 PM, 18 August 2018

Question # 1 of 10 (Start time: 07:48:23 PM, 18 August 2018) Total Marks: 1

Let

$$f : G \rightarrow H$$

be a homomorphism, then $f(e_G) = \text{---}$ for e_G and e_H are identities of G and H respectively.

Select the correct option

Reload Math Equations

☐	e_G
☒	e_H

Click to Save Answer & Move to Next Question

AHMAD MSC(MATH)
Areeb2113@gmail.com

Question # 7 of 10 (Start time: 07:50:22 PM, 18 August 2018)

Total Marks: 1

Let φ be a homomorphism function from group $(G, *)$ to group $(G', \cdot)$. Then, $(\text{Ker } \varphi, *)$ is a normal subgroup of $(G, *)$.

Select the correct option

Reload Math Equations

True

False

Click to Save Answer & Move to Next Question

$\mathbb{R}$ with usual topology is a T_4

Select the correct option

☒	True
☐	False

Question # 9 of 10 (Start time: 08:44:38 PM, 01 February 2020)

Every closed subspace of a compact space is compact.

Select the correct option

☒	True
☐	False

Question # 8 of 10 (Start time: 08:43:43 PM, 01 February 2020)

A set X with cofinite topology is compact.

Select the correct option

☒	True
☐	False

MC180404646: AMINA SHAHZADI

MTH634:Quiz No.2

Question # 7 of 10 (Start time: 08:42:31 PM, 01 February 2020)

All the metric spaces are not normal spaces.

Select the correct option

☐	True
☒	False

If (X, τ) be a compact Hausdorff space then

Select the correct option

☐	True
☒	False

Question # 5 of 10 (Start time: 08:40:57 PM, 01 February 2020)

Image of a compact space under a continuous map is compact.

Select the correct option

☒	True
☐	False

MC180404646: AMINA SHAHZADI

MTH634:Quiz No.2

Question # 4 of 10 (Start time: 08:40:15 PM, 01 February 2020)

Every compact subspace of a Hausdorff space is not closed.

Select the correct option

☐	True
☒	False

Consider $\mathbb{R}$ with usual topology. There exists no homomorphism between

Select the correct option

☒	True
☐	False

MC180404646: AMINA SHAHZADI

MTH634:Quiz No.2

Question # 2 of 10 (Start time: 08:38:48 PM, 01 February 2020)

Which of the following statement is true?

Select the correct option

☐	$\mathbb{R}$ with usual topology is compact
☒	$\mathbb{R}$ with indiscrete topology is compact
☐	$\mathbb{R}$ with discrete topology is compact
☐	$\mathbb{R}$ with cofinite topology is not compact

If (X, τ) be a compact Hausdorff space then

Select the correct option

☐	True
☒	False

MC180404844: RABIA SHAFEE

MTH634:Quiz No.2

Question # 9 of 10 (Start time: 08:08:12 PM, 01 February 2020)

All the metric spaces are not normal spaces.

Select the correct option

☐	True
☒	False

MC180404844: RABIA SHAFEE

MTH634:Quiz No.2

Question # 8 of 10 (Start time: 08:07:30 PM, 01 February 2020)

Every closed subspace of a compact space is compact.

Select the correct option

☒	True
☐	False

MC180404844: RABIA SHAFEE

MTH634:Quiz No.2

Question # 7 of 10 (Start time: 08:06:53 PM, 01 February 2020)

Every discrete space is not a regular space.

Select the correct option

☐	True
☒	False

Question # 6 of 10 (Start time: 08:06:13 PM, 01 February 2020)

An open interval in

$\mathbb{R}$

with usual topology is not compact.

Select the correct option

☒	True
☐	False

MC180404844: RABIA SHAFEE

MTH634:Quiz No.2

Question # 5 of 10 (Start time: 08:05:23 PM, 01 February 2020)

Image of a compact space under a continuous map is compact.

Select the correct option

☒	True
☐	False

MC180404844: RABIA SHAFEE

MTH634:Quiz No.2

Question # 4 of 10 (Start time: 08:04:16 PM, 01 February 2020)

$\mathbb{R}$ with usual topology is a T_4

Select the correct option

☒	True
☐	False

MC180404844: RABIA SHAFEE

MTH634:Quiz No.2

Question # 3 of 10 (Start time: 08:03:32 PM, 01 February 2020)

Every compact subspace of a Hausdorff space is not closed.

Select the correct option

☐	True
☒	False

MC180404844: RABIA SHAFEE

MTH634:Quiz No.2

Question # 2 of 10 (Start time: 08:02:50 PM, 01 February 2020)

Consider $\mathbb{R}$ with usual topology. There exists no homomorphism between

Select the correct option

☒	True
☐	False

Question # 1 of 10 (Start time: 08:02:17 PM, 01 February 2020)

Which of the following statement is false?

Select the correct option

☒	Every metric space is second countable.
☐	Every metric space is Hausdorff space.
☐	Every metric space is normal space.
☐	none

MC180403576: SHAZIA HANIF

MTH634:Quiz No.2

Question # 10 of 10 (Start time: 08:30:14 PM, 01 February 2020)

$\mathbb{R}$ with usual topology is a T_4

Select the correct option

☒	True
☐	False

Consider $\mathbb{R}$ with usual topology. There exists no homomorphism between

Select the correct option

☒	True
☐	False

Question # 8 of 10 (Start time: 08:28:16 PM, 01 February 2020)

Which of the following statement is false?

Select the correct option

☐	A set X with indiscrete topology is compact.
☐	A set X with any topology containing finite number subsets of X is compact.
☐	A finite set X with any topology is compact.
☒	An infinite set X with discrete topology is compact.

Question # 6 of 10 (Start time: 08:26:59 PM, 01 February 2020)

Which of the following statement is true?

Select the correct option

☐	$\mathbb{R}$ with usual topology is compact
☒	$\mathbb{R}$ with indiscrete topology is compact
☐	$\mathbb{R}$ with discrete topology is compact
☐	$\mathbb{R}$ with cofinite topology is not compact

Question # 5 of 10 (Start time: 08:26:20 PM, 01 February 2020)

A set X with cofinite topology is compact.

Select the correct option

☒	True
☐	False

If (X, τ) be a compact Hausdorff space then

Select the correct option

☐	True
☒	False

Question # 3 of 10 (Start time: 08:24:58 PM, 01 February 2020)

Which of the following statement is false?

Select the correct option

☒	Every metric space is second countable.
☐	Every metric space is Hausdorff space.
☐	Every metric space is normal space.
☐	none

Question # 2 of 10 (Start time: 08:23:35 PM, 01 February 2020)

Every closed subspace of a compact space is compact.

Select the correct option

☒	True
☐	False

Question # 1 of 10 (Start time: 08:38:02 PM, 01 February 2020)

Which of the following statement is false?

Select the correct option

☐	A set X with indiscrete topology is compact.
☐	A set X with any topology containing finite number subsets of X is compact.
☐	A finite set X with any topology is compact.
☒	An infinite set X with discrete topology is compact.